1). Create Departments Table

```
CREATE TABLE departments
(
 id INT IDENTITY (1,1) PRIMARY KEY,
 name VARCHAR (50)
);

CREATE TABLE employees
(
 id INT IDENTITY (1,1) PRIMARY KEY,
 name VARCHAR (50),
 department_id INT,
 Salary DECIMAL (10,2),
 manager_id INT,
 Joining_date DATE,
FOREIGN KEY (department_id) REFERENCE departments id,
FOREIGN KEY (manager_id) REFERENCE employees_id,
);

CREATE TABLE projects
(
```

```
id INT IDENTITY(1,1) PRIMARY KEY,
name VARCHAR (50),
start_date DATE,
end_date DATE.
);

CREATE TABLE employee_Projects
(
employee_id INT,
project_id INT,
allocated_date DATE,
PRIMARY KEY (employee_id, project_id),
FOREIGN KEY (employee_id) REFERENCES employees(id),
FOREIGN KEY (project_id) REFERENCES projects(id)
);

CREATE UNIQUE INDEX UQ_DepartmentName
ON departments (name);

ALTER TABLE employees
ADD email VARCHAR (100);

SELECT e.name As Employee, m.name As Manager
FROM employees e
```

```
LEFT JOIN employees m
ON e.manager_id = m.id;

SELECT department_id,
       SUM(salary) AS Total salary_
       AVG(salary) AS Average salary_
       COUNT(*) AS Employee Count
FROM employees
GROUP BY department_id
HAVING COUNT(*) >5;

SELECT name, salary
FROM employees e
WHERE salary >
(
  SELECT AVG(salary)
  FROM employees
  WHERE department_id = e.department_id
);
```

WIT
(

```
WITH EmpRank As
(
 SELECT name,
     department_id,
     salary,
     Row() NUMBER() OVER (PARITION BY
       department_id ORDER BY salary DESC)
        AS RankNo

 FROM Employees
)
SELECT name, department_id, salary
FROM EmpRank
 WHERE RankNO <= 2;
```